

Week 2 — Survey Data Sources & Reading a Methods Section Critically

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Week 2

Survey Data Sources & Reading a Methods Section Critically

The data engine underneath the paper — and how a replicator reads Methods.

- ▶ Data source: **NHANES** (public, documented, versioned)
- ▶ Skill: strip a Methods section for its reproducible parts
- ▶ Feeds **L1** (Thu) and **M0** shortlisting (due Wk3)

Learning objectives

- ▶ Describe **NHANES structure**: 2-year cycles; interview / MEC / lab components; the SEQN merge key; the fasting subsample
- ▶ Explain why MASLD is a **surrogate-defined** outcome here (**Fatty Liver Index**) — what non-invasive scores buy and cost
- ▶ Read a Methods section to extract the **four primitives**: exposure, outcome, eligibility, variables
- ▶ State the **six M0 feasibility criteria** — and place the four primitives correctly inside them: reconstructability is the **pre-screen**, not the whole gate
- ▶ Anticipate **L1**: download multiple NHANES cycles and merge on SEQN

Where we've been, where we're going

- ▶ **Wk1 recap:** we set up the course and only **named** the paper (Kueh 2026, MASLD phenotypes & mortality) — one teaser slide, no methods
- ▶ We unpack the paper **one piece at a time**, the week each method needs it — **today is the first piece**
- ▶ **Today:** open the *data engine* (**NHANES**) + read Methods like a replicator — and meet **what MASLD is** and the **Fatty Liver Index**
- ▶ **Sets up:** Thu **L1** (multi-cycle download + merge on SEQN) and **M0** (shortlist candidate papers)

Where we are in the project

- ▶ Term map: **Replicate (Wk2–8) → Interrogate → Improve → Defend**
- ▶ Milestones: **M0 (Wk3)** through **M4 (Dec)**
- ▶ Today's job: understand the **data source** and *strip a Methods section for parts*
- ▶ Two deliverables this week feeds:
 - ▶ **L1** (Thu lab)
 - ▶ **M0 shortlisting** (due Wk3)

Why start with the data source

- ▶ You cannot judge or reproduce a study you don't understand **at the data level**
- ▶ The Methods section is a **promise** about data — one you must be able to keep
- ▶ NHANES is **public, documented, and versioned** — ideal for a replication course

MASLD: Kueh 2026 uses NHANES **2007–2018** (six cycles) — every number we reproduce traces back to public files.

NHANES in one picture

- ▶ Continuous NHANES released in **2-year cycles**: 2007-08, 2009-10, ... 2017-18
- ▶ ~5,000 examined persons per cycle after screening
- ▶ Nationally representative of the US civilian non-institutionalized population via a **complex multistage design** (weights exist — we *use* them in Wk6)
- ▶ Data arrive as **many small files by topic**, not one table

MASLD: “2007–2018” = six cycles stacked; combining cycles is why the analytic **N runs to several thousand** — the paper reports ~**6,300**, our **full analytic file** lands on **6,371**.

The three data-collection settings

- ▶ **Interview (questionnaire)** — at home: demographics, income, alcohol, medical history
- ▶ **MEC exam** (Mobile Examination Center) — physical measures: **BMI**, **waist circumference**, blood pressure
- ▶ **Laboratory** — blood draws: **triglycerides**, **GGT**, glucose; some assays only in the **fasting subsample**

MASLD: FLI needs triglycerides + GGT (lab) + BMI + waist (MEC) — the exposure spans *two* settings, so anyone missing either is **missing the exposure**.

SEQN: the key that holds it together

- ▶ Every respondent has one **SEQN** (respondent sequence number), unique *within* a cycle
- ▶ Analysis = pick the topic files you need, then **merge on SEQN** into one row-per-person table
- ▶ **Gotcha:** SEQN is unique within a cycle, *not across* cycles → add a **cycle indicator** before stacking

MASLD: reproducing Table 1 means merging DEMO + BMX + BIOPRO + TRIGLY + ALQ + mortality — all keyed on SEQN, across six cycles.

The fasting subsample (and why it will haunt us)

- ▶ Some labs (fasting glucose, triglycerides on the calculation path) come **only** from the morning **fasting subsample** → smaller N, its own weight (WTSAF)
- ▶ Being in the fasting subsample is **not random** with respect to health — a preview of **selection bias**

MASLD: FLI needs fasting labs, so it is computable only in NHANES' **designed fasting subsample**. That is a *design* feature with its own weight, **not** missing data.

What “MASLD” actually is

- ▶ **MASLD** = metabolic dysfunction-associated steatotic liver disease (the 2023 rename of **NAFLD**): hepatic **steatosis** + 1 cardiometabolic criterion
- ▶ Gold-standard steatosis = **biopsy or imaging** — neither exists in NHANES for this window
- ▶ So the paper needs a **surrogate** for “fatty liver”

MASLD: the disease definition itself *forces* a non-invasive workaround — a feature of the data, not a shortcut by the authors.

Surrogate outcomes: the Fatty Liver Index

- ▶ **FLI** is a published score from **triglycerides, BMI, GGT, waist circumference**
- ▶ Kueh calls steatosis at **FLI 60**
- ▶ Non-invasive scores make population studies possible — but **embed their inputs into the case definition**

MASLD: seed for **M1** — the phenotype groups are built from **BMI × waist (WHtR)**, and FLI eligibility *also* uses **BMI + waist**. Exposure and cohort selection share inputs → **circularity**. Name it today; dissect it at M1.

Break

10 minutes.

Back for the second half: the **four primitives** and the **M0 feasibility** test.

The four primitives of a reproducible study

- ▶ To reproduce *any* paper, extract four things from the Methods:
 - ▶ **Exposure**
 - ▶ **Outcome**
 - ▶ **Eligibility**
 - ▶ **Variables** (confounders / covariates)
- ▶ If any one is under-specified, the study is **not reconstructable** — no matter how clean the results look

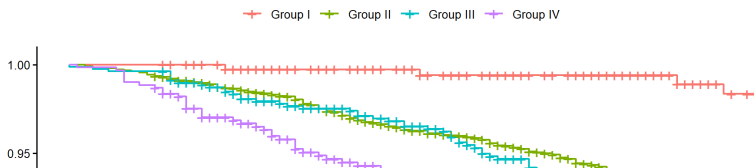
MASLD: we'll find **three** boxes clear and **one** (eligibility) *internally contradictory*.

Extract the EXPOSURE

- ▶ Exposure = **four phenotypes** from obesity (**BMI** 30) $\times$ central adiposity (**WHtR** 0.6):
 - ▶ **I** obese / low-central
 - ▶ **II** obese / high-central
 - ▶ **III** non-obese / low-central
 - ▶ **IV** non-obese / high-central
- ▶ Ask: are cut-points, variables, and coding fully stated?
Mostly yes.

MASLD: the **paper's** group sizes are **386 / 4,541 / 769 / 604**; **our reproduction** gives **391 / 4,585 / 787 / 608**. Either way **Group IV** (non-obese, high-central) has the **worst survival** — the paper's headline.

All-Cause Mortality



Extract the OUTCOME and follow-up

- ▶ Outcome = **all-cause mortality** via NHANES **linked mortality files**
- ▶ Time-to-event; **median follow-up ~6.7 y**
- ▶ Ask: mortality source, censoring date, time origin all stated?
Yes — linked NDI is standard and documented

MASLD: the paper counts **585** all-cause deaths, our reproduction **586**; Group IV **unadjusted HR ~15.1** collapses to **~2.9** after adjustment — a first lesson that **confounding is enormous** here (Wk4 preview).

Extract ELIGIBILITY — read it adversarially

- ▶ Eligibility = adults with **MASLD** (FLI $\geq 60 + 1$ cardiometabolic criterion), plus stated **alcohol** and **missingness** exclusions
- ▶ Reading critically = check that the prose **arithmetic** actually lands on the reported **N**

MASLD — two real discrepancies on screen: {.incremental}

- ▶ Alcohol-exclusion sentence, coded three plausible ways $\rightarrow$ **N** = **4,308 / 5,944 / 6,371** — the text doesn't pin down the cohort
- ▶ Table headers mislabel the BMI cut as “**<25**” when the analysis uses **30** — never trust a label you haven't recomputed

Extract the VARIABLES (confounders & covariates)

- ▶ Variables = **age, sex, race/ethnicity, income, smoking, comorbidities** — each must map to a specific NHANES file/field
- ▶ Ask: is every adjustment variable named precisely enough to pull it? **Mostly** — but **income has ~9% missing** (a Wk7 thread)

MASLD: reproduced Table 1 matches closely (**mean age ~51, BMI ~33.2**) — evidence the variable definitions are recoverable even where the eligibility prose is not.

M0 feasibility: the four boxes are a pre-screen

A paper is **reconstructable** only if a replicator can — from public data + Methods alone — fill all four boxes:

- ▶ **Data:** public, obtainable source (ideally NHANES) with stated version/cycles
- ▶ **Exposure & outcome:** explicit variables and cut-points
- ▶ **Eligibility:** resolves to a **countable N** (flag contradictions early — they're *interrogation fuel*, not disqualifiers)
- ▶ **Confounders:** named, mapping to real fields

*A little ambiguity is fine — it's what M1 interrogates. **Total** under-specification is not.*

Necessary, not sufficient: filling the boxes makes a paper reconstructable. It does not get it through **M0**.

M0 has six criteria — the boxes reach three

The boxes give you criteria **1–3**: reconstructable data, a clear exposure and outcome, plausible confounders. Three more must **also** hold:

- ▶ **4 · Size and events** — enough observations, and for time-to-event outcomes enough events, for the main analysis *and* at least one stable subgroup/interaction analysis
- ▶ **5 · Design variables** — the design feature the analysis has to respect, and what encodes it: survey weights/strata/PSU, a cluster or repeated-measure ID, or follow-up time and censoring. **Not waivable**
- ▶ **6 · A genuine incomplete-observation problem** — a variable the locked analysis *actually uses* that is absent for records which should have carried it. Shrinkage from designed subsampling, eligibility or the disease definition is **not** this

MASLD: Kueh 2026 clears **all six** — which is why it can carry P1–P5 — *and* hands us a rich interrogation agenda.

Common traps when reading Methods

- ▶ **Labels vs. computations** — the “ $<25 = 30$ ” trap
- ▶ **Prose arithmetic** that never sums to the reported N
- ▶ **Silent design choices** — unweighted analysis of a survey; a fasting-only exposure treated as universal
- ▶ **“Defined-by” scores** whose inputs overlap the exposure → **circularity** — log it now, test at M1

MASLD: every trap on this slide is one the demonstration paper actually contains.

Bridge to today's lab (L1) & your increment

- ▶ **In L1 (Thu, EpiMethods wrangling module):** download several NHANES cycles (**2007–2018**), read raw topic files (DEMO, BMX, lab files), and **merge on SEQN** into one analytic-ready, one-row-per-person table — adding a **cycle indicator before stacking**. *Output: a merged multi-cycle dataset you can count and describe.*
- ▶ **Why it matters:** L1's merge is the **substrate for the whole term** — every later lab (Table 1, confounding, interaction, weights, missing data) runs on this table
- ▶ **M0 shortlisting (toward Wk3 lock):** using today's **four-primitive template** + the **six M0 criteria**, shortlist **2–3 candidate NHANES papers**; for each, be ready to say whether exposure / outcome / eligibility / variables are extractable — the **pre-screen** — *and* whether the paper has enough observations and events, a **design feature to respect**, and a **missing variable the analysis actually uses**. **Lock one at M0 (Wk3).**

Takeaway: reading a Methods section is the first act of